

# SQL Syllabus

Course Duration: 40 hrs.

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| 1 | <b>Database</b> <ul style="list-style-type: none"><li>• What is Database or DBMS</li><li>• What is RDBMS</li><li>• Types of SQL Databases</li><li>• Database Normalization</li><li>• Types of Normalization</li><li>• Installing MySQL and Configurations</li></ul>                                                          |
| 2 | <b>Java MySQL Database Access</b> <ul style="list-style-type: none"><li>• Database Handling with MySQL</li><li>• Create Database Connection</li><li>• <b>SQL Constrains</b><ul style="list-style-type: none"><li>• Primary Key</li><li>• Check</li><li>• Not Null</li><li>• Foreign Key</li><li>• Unique</li></ul></li></ul> |
| 3 | <b>DDL Operations</b> <ul style="list-style-type: none"><li>• What is DDL command</li><li>• Creating database, Table</li><li>• Dropping Database, Table</li><li>• Altering the table</li><li>• Using Truncate</li></ul>                                                                                                      |

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| 4 | <b>DML Operations</b> <ul style="list-style-type: none"><li>• What is DML</li><li>• Selecting the row or column</li><li>• Inserting data into table</li><li>• Updating the data in table</li><li>• Deleting data or row, columns</li></ul> |
| 5 | <b>Reading Data in SQL</b> <ul style="list-style-type: none"><li>• Distinct</li><li>• Limit</li><li>• Offset</li><li>• Fetch</li><li>• Case</li></ul>                                                                                      |
| 6 | <b>SQL DataTypes</b><br><b>SQL Operators</b> <ul style="list-style-type: none"><li>• Arithmetic</li><li>• Relational</li><li>• Logical</li></ul>                                                                                           |
| 7 | <b>SQL Expression</b> <ul style="list-style-type: none"><li>• Boolean</li><li>• Numeric</li><li>• Date</li></ul>                                                                                                                           |

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| 8  | <b>Advanced SQL Topics</b><br><b>Filtering</b> <ul style="list-style-type: none"><li>• Where Clause</li><li>• Like</li><li>• IN</li><li>• Between</li><li>• Order By</li><li>• Group By</li><li>• Is Null</li></ul>                                  |
| 9  | <b>Aggregation in SQL</b> <ul style="list-style-type: none"><li>• Count</li><li>• Sum</li><li>• Avg</li><li>• Min</li><li>• Max</li><li>• Having</li></ul>                                                                                           |
| 10 | <b>SQL Function</b> <ul style="list-style-type: none"><li>• Scalar Function - Upper()</li><li>• Aggregate Function - AVG()</li><li>• String Function – CONCAT(), SUBSTR()</li><li>• Date and Time Function</li><li>• Mathematical Function</li></ul> |

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| 11 | <b>SQL Joins</b> <ul style="list-style-type: none"><li>• What is Joins</li><li>• Inner Join</li><li>• Outer Join<ol style="list-style-type: none"><li>1. Left Join</li><li>2. Right Join</li><li>3. Full Join</li></ol></li><li>• Cross Join</li><li>• Self-Join</li><li>• Union</li></ul> |
| 12 | <b>Subqueries in SQL</b> <ul style="list-style-type: none"><li>• Single-row Subquery</li><li>• Multiple-row Subquery</li><li>• Correlated Subquery</li><li>• Nested Subquery</li></ul>                                                                                                     |
| 13 | <b>Views in SQL</b><br><b>Index in SQL</b> <ul style="list-style-type: none"><li>• Create</li><li>• Drop</li></ul>                                                                                                                                                                         |
| 14 | <b>Exercise &amp; Practice</b>                                                                                                                                                                                                                                                             |